```
proc import out=mall
    datafile="C:\Users\kawao\Documents\CSULB\2025 Spring\STAT 560
\Project\Mall_Customers.csv"
    dbms=csv
    replace;
    getnames=yes;
run;
/* Spending_Score is from 1-100*/
/* Annual_Income is in k$*/

title "EDA for Gender";
/* ANOVA with residuals and predicted values */
proc glm data=mall;
    class Gender;
    model Spending_Score = Gender;
    output out=DiagOutGender r=Residual p=Predicted;
run;
quit;

/* Normality of residuals */
proc univariate data=DiagOutGender normal;
    var Residual;
    histogram Residual / normal;
    qqplot Residual / normal(mu=est sigma=est);
run;

/* Residuals vs. predicted plot */
proc sgplot data=DiagOutGender;
    scatter x=Predicted y=Residual;
    refline 0 / axis=y lineattrs=(color=red);
    title "Residuals vs. Predicted Values";
run;

/* Homogeneity of variances */
proc glm data=mall;
    class Gender;
    model Spending_Score = Gender;
    means Gender / hovtest=levene;
run;
quit;

title "Do the spending scores differ across both genders?";
proc glm data=mall;
    class Gender;
    model Spending_Score = Gender;
run;

/* Wilcoxon Rank Sum Test */
proc npar1way data=mall wilcoxon;
    class Gender;
    var Spending_Score;
run;
```

```
title "EDA for Age_Group";
data mall;
    set mall;
    if Age < 25 then Age_Group = "Under 25";
    else if Age < 40 then Age_Group = "25-39";
    else if Age < 60 then Age_Group = "40-59";
    else Age_Group = "60+";
run;

/* Run ANOVA with residuals and predicted values */
proc glm data=mall;
    class Age_Group;
    model Spending_Score = Age_Group;
    output out=DiagOutAge r=Residual p=Predicted;
run;
quit;

/* Normality of residuals */
proc univariate data=DiagOutAge normal;
    var Residual;
    histogram Residual / normal;
    qqplot Residual / normal(mu=est sigma=est);
run;

/* Residuals vs. Predicted Plot */
proc sgplot data=DiagOutAge;
    scatter x=Predicted y=Residual;
    refline 0 / axis=y lineattrs=(color=red);
    title "Residuals vs. Predicted Values";
run;

/* Homogeneity of variances */
proc glm data=mall;
    class Age_Group;
    model Spending_Score = Age_Group;
    means Age_Group / hovtest=levene;
run;
quit;

title "Do the annual spendings differ across age groups?";
proc glm data=mall;
    class Age_Group;
    model Spending_Score = Age_Group;
run;

/* Kruskal-Wallis Test*/
proc npar1way data=mall wilcoxon;
    class Age_Group;
    var Spending_Score;
run;
```

```
title "Wilcoxon Rank Sum Test between the age groups";
/* Under 25 vs 25-39 */
proc npar1way data=mall wilcoxon;
    where Age_Group in ("Under 25", "25-39");
    class Age_Group;
    var Spending_Score;
    title "Wilcoxon: Under 25 vs 25-39";
run;

/* Under 25 vs 40-59 */
proc npar1way data=mall wilcoxon;
    where Age_Group in ("Under 25", "40-59");
    class Age_Group;
    var Spending_Score;
    title "Wilcoxon: Under 25 vs 40-59";
run;

/* Under 25 vs 60+ */
proc npar1way data=mall wilcoxon;
    where Age_Group in ("Under 25", "60+");
    class Age_Group;
    var Spending_Score;
    title "Wilcoxon: Under 25 vs 60+";
run;

/* 25-39 vs 40-59 */
proc npar1way data=mall wilcoxon;
    where Age_Group in ("25-39", "40-59");
    class Age_Group;
    var Spending_Score;
    title "Wilcoxon: 25-39 vs 40-59";
run;

/* 25-39 vs 60+ */
proc npar1way data=mall wilcoxon;
    where Age_Group in ("25-39", "60+");
    class Age_Group;
    var Spending_Score;
    title "Wilcoxon: 25-39 vs 60+";
run;

/* 40-59 vs 60+ */
proc npar1way data=mall wilcoxon;
    where Age_Group in ("40-59", "60+");
    class Age_Group;
    var Spending_Score;
```

```
    title "Wilcoxon: 40-59 vs 60+";
run;


title "ANOVA between age groups";
proc glm data=mall;
    class Age_Group;
    model Spending_Score = Age_Group;
    means Age_Group / tukey cldiff alpha=0.05;
    title "Tukey HSD Post-Hoc for Age_Group ANOVA";
run;
quit;
```